

# An explicit counterexample to the Special Image Conjecture in dimension 21

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with heavy assistance from ChatGPT 5.6 Sol

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## Abstract

Let

$$E : \mathbb{C}[\xi_1, \dots, \xi_n, Z_1, \dots, Z_n] \longrightarrow \mathbb{C}[Z], \quad E(\xi^\alpha p(Z)) = \partial_Z^\alpha p(Z).$$

The Special Image Conjecture  $\text{SIC}(n)$  says that  $\ker E$  is a Mathieu–Zhao subspace. We give an exact rational polynomial  $A \in \mathbb{Q}[\xi, Z]$  for  $n = 21$ , with 72 monomials and total degree four, and take  $b = Z_1$ . We prove

$$A^m \in \ker E \quad (m \geq 1), \quad bA^m \notin \ker E \quad \text{for infinitely many } m.$$

Thus  $\text{SIC}(21)$  is false. The witness is obtained from a certified 13-variable stable model of the newly announced Keller counterexample by adjoining eight variables. The bridge is a scalar-parameter form of Abhyankar–Gurjar inversion; unlike Zhao’s homogeneous specialization, it permits the linear block in this construction. Exact SymPy and dependency-free rational checkers accompany a 72-term sparse certificate.

**Verification disclaimer.** The human author is a complete amateur exploring the limits of AI-assisted mathematics and cannot independently verify these claims. This is a research artifact for expert checking, not an established result. Priority is provisional and limited by the documented audit.

## 1 The Special Image Conjecture

Put  $R_n = \mathbb{C}[\xi_1, \dots, \xi_n, Z_1, \dots, Z_n]$  and define  $E_n : R_n \rightarrow \mathbb{C}[Z]$  by

$$E_n(\xi^\alpha p(Z)) = \partial_Z^\alpha p(Z).$$

If  $\Theta_j = \xi_j - \partial_{Z_j}$ , Zhao’s theorem gives

$$\ker E_n = \sum_{j=1}^n \Theta_j R_n. \tag{1}$$

A linear subspace  $M$  of a commutative algebra is a Mathieu–Zhao subspace if

$$f^m \in M \quad (m \geq 1) \implies qf^m \in M \quad (m \gg 0) \quad \text{for every } q.$$

The assertion that  $\ker E_n$  is such a subspace is  $\text{SIC}(n)$ .

## 2 A scalar-parameter inversion lemma

The following form is useful because it does not require  $g$  to be homogeneous or to have order at least two.

**Lemma 1** (pencil-Keller collision lemma). *Let  $k$  have characteristic zero, let  $g \in k[Z]^n$ , and suppose*

$$\det(I + tJg) = 1 \quad \text{in } k[t, Z]. \quad (2)$$

Set

$$A(\xi, Z) = - \sum_{j=1}^n \xi_j g_j(Z).$$

Then  $E_n(A^m) = 0$  for every  $m \geq 1$ . If  $T = I + g$  has two points  $p, q$  with  $T(p) = T(q)$  and  $p_i \neq q_i$ , then

$$E_n(Z_i A^m) \neq 0$$

for infinitely many  $m$ .

*Proof.* Work in the  $t$ -adically complete ring  $k[Z][[t]]$ . The map

$$T_t(Z) = Z + tg(Z) = Z - H_t(Z), \quad H_t = -tg,$$

has a unique formal inverse  $Q_t$ , because it is congruent to the identity modulo  $t$ . The Abhyankar–Gurjar inversion identity is valid in this filtration: for  $p \in k[Z]$ ,

$$p(Q_t(Z)) = \sum_{\alpha \in \mathbb{N}^n} \frac{1}{\alpha!} \partial_Z^\alpha (H_t(Z)^\alpha \det JT_t(Z) p(Z)). \quad (3)$$

Indeed, apply the usual formal identity modulo  $t^{N+1}$  for every  $N$ ;  $H_t^\alpha$  has  $t$ -adic order  $|\alpha|$ , so every coefficient is a finite sum. By (2),  $\det JT_t = 1$ . The multinomial theorem turns (3) into

$$p(Q_t(Z)) = \sum_{m \geq 0} \frac{t^m}{m!} E_n(p(Z) A^m). \quad (4)$$

Taking  $p = 1$  makes the left side equal to one, proving  $E_n(A^m) = 0$  for every positive  $m$ .

Now take  $p = Z_i$ . If  $E_n(Z_i A^m) = 0$  for all sufficiently large  $m$ , (4) says that the  $i$ th coordinate of  $Q_t$  is a polynomial in  $t, Z$ . The formal identity  $Q_{t,i}(T_t(Z)) = Z_i$  is then a polynomial identity and may be specialized at  $t = 1$ . Hence  $Q_{1,i}(T(p)) = p_i$  and  $Q_{1,i}(T(q)) = q_i$ , contradicting  $T(p) = T(q)$  and  $p_i \neq q_i$ .  $\square$

The  $t$ -adic formulation matters below:  $g$  has a linear  $BU$  block. Zhao’s homogeneous theorem should therefore not be quoted verbatim for this witness; Lemma 1 supplies the needed extension.

## 3 The explicit 21-variable map

Use the coordinate order

$$Z = (x, y, z, a_1, b_1, a_2, b_2, a_3, b_3, a_4, b_4, a_5, b_5, U_1, \dots, U_8).$$

Define  $g = (g_1, \dots, g_{21})$  as follows.

$$\begin{aligned}
g_1 &= U_1 + \frac{1}{2}a_1b_1 + \frac{3}{2}a_1y, & g_2 &= U_2 - a_2b_2 - 2a_2z + a_3b_3 + 3xz, \\
g_3 &= U_3 + b_1a_5 - a_3b_6 - a_4b_4 - 7a_4y + 3a_5y + 4y^2, & g_4 &= x^2, \\
g_5 &= -3U_2 + 3a_2b_2 + 6a_2z - 3a_3b_3 - 8xz, & g_6 &= U_4, \\
g_7 &= U_5 - 2b_1a_5 + 2a_3b_6 + 2a_4b_4 + 14a_4y - 6a_5y - 5y^2, & g_8 &= xy, \\
g_9 &= a_2z + 3b_2x, & g_{10} &= U_6, \\
g_{11} &= U_7 - b_1a_3 + 7a_2b_2 + 14a_2z - 7a_3b_3 - 3a_3y - 18xz, & g_{12} &= U_8, \\
g_{13} &= b_1a_4 - b_4y.
\end{aligned} \tag{5}$$

$$\begin{aligned}
g_{14} &= -\frac{1}{2}a_1xz - \frac{1}{2}b_1x^2, \\
g_{15} &= -a_2a_3z + 3a_2y^2 - 3b_2a_3x - b_3xy - 12xy^2, \\
g_{16} &= b_1a_3a_4 - a_3b_4y + 3a_4xz - a_5xz + b_6xy - 3xyz, \\
g_{17} &= -3x^2y, \\
g_{18} &= -2b_1a_3a_4 + 2a_3b_4y - 6a_4xz + 2a_5xz - 2b_6xy + 5xyz, \\
g_{19} &= -xy^2, \\
g_{20} &= b_1xy + 7a_2a_3z - 21a_2y^2 + 21b_2a_3x + a_3xz + 7b_3xy + 84xy^2, \\
g_{21} &= -a_4xy.
\end{aligned} \tag{6}$$

The polynomial in the theorem is

$$A(\xi, Z) = -\sum_{j=1}^{21} \xi_j g_j(Z), \quad b = Z_1 = x. \tag{7}$$

It has exactly 72 monomials and total degree four. Equations (5)–(7) are a human-readable certificate; the accompanying JSON records the same object as exact sparse rational data.

## 4 Determinant certificate

The structure behind the display is an exactly certified 13-variable stable model from Exploration 03,

$$\Psi(X) = X + H_2(X) + BK(X), \tag{8}$$

where  $H_2$  is quadratic,  $K$  has eight cubic components, and  $B \in \text{Mat}_{13 \times 8}(\mathbb{Q})$ . Its Jacobian determinant is one. With  $Z = (X, U)$ , the displayed vector is

$$g(X, U) = (H_2(X) + BU, -K(X)). \tag{9}$$

Therefore

$$I + sJg = \begin{pmatrix} I_{13} + sJH_2 & sB \\ -sJK & I_8 \end{pmatrix}.$$

Taking the Schur complement of  $I_8$ , and using the homogeneities of  $H_2$  and  $K$ , gives the polynomial identity

$$\begin{aligned}
\det(I + sJg) &= \det(I_{13} + sJH_2 + s^2BJK) \\
&= \det J\Psi(sX) = 1.
\end{aligned} \tag{10}$$

The primary checker verifies the matrix equality in (10) entry by entry. The dependency-free checker separately tests the determinant pencil at 66 exact rational specializations; a Node.js/BigInt implementation supplies a third runtime and checks 18 further exact specializations.

## 5 Collision certificate and theorem

The following three rational points use the coordinate order above.

$$\begin{aligned} p_0 &= (0, 0, -\frac{1}{4}, 0, 0, 0, \frac{1}{2}, 0, 0, 0, 0, 0, 0; 0, 0, 0, 0, 0, 0, 0, 0), \\ p_+ &= (1, -\frac{3}{2}, \frac{13}{2}, -1, -2, \frac{9}{2}, -10, \frac{3}{2}, \frac{3}{4}, -\frac{9}{4}, -6, -\frac{27}{8}, \frac{9}{2}; \\ &\quad -\frac{17}{4}, -\frac{45}{8}, \frac{99}{16}, -\frac{9}{2}, -\frac{177}{8}, \frac{9}{4}, \frac{213}{8}, \frac{27}{8}), \\ p_- &= (-1, \frac{3}{2}, \frac{13}{2}, -1, 2, -\frac{9}{2}, -10, \frac{3}{2}, -\frac{3}{4}, \frac{9}{4}, 6, \frac{27}{8}, \frac{9}{2}; \\ &\quad \frac{17}{4}, \frac{45}{8}, \frac{99}{16}, \frac{9}{2}, -\frac{177}{8}, -\frac{9}{4}, -\frac{213}{8}, -\frac{27}{8}). \end{aligned}$$

Direct substitution gives

$$(I + g)(p_0) = (I + g)(p_+) = (I + g)(p_-) = (0, 0, -\frac{1}{4}, 0, 0, 0, \frac{1}{2}, 0, \dots, 0). \quad (11)$$

Their first coordinates are  $0, 1, -1$ .

**Theorem 2.** *For  $A$  and  $b$  in (7),*

$$A^m \in \ker E_{21} \quad \text{for every } m \geq 1,$$

*but*

$$bA^m \notin \ker E_{21}$$

for infinitely many  $m$ . Consequently SIC(21) is false.

*Proof.* Equation (10) and the collision  $p_0, p_+$  in (11), separated by their first coordinate, satisfy Lemma 1 with  $n = 21$  and  $i = 1$ . The two assertions follow. Equation (1) identifies  $\ker E_{21}$  with the image appearing in the usual statement of the Special Image Conjecture.  $\square$

As low-order guards, the symbolic checker also obtains

$$E_{21}(A) = E_{21}(A^2) = 0, \quad E_{21}(bA) \neq 0, \quad E_{21}(bA^2) = 3x^2y \neq 0. \quad (12)$$

## 6 Verification and scope

The repository contains three exact implementations. The SymPy checker reconstructs the stable model, verifies the Schur-complement matrix identity, the 72-term count, the collision, and (12). A dependency-free checker parses only the exported JSON, rebuilds  $A = -\xi \cdot g$ , evaluates the collision with rational arithmetic, and checks the determinant pencil at 66 exact specializations. A separate Node.js/BigInt implementation rechecks the collision and 18 determinant specializations without Python or a computer algebra system.

Zhang’s immediate-consequences note correctly established only that the Image Conjecture fails in some finite dimension. There is nevertheless an immediate explicit predecessor: Exploration 03’s 22-variable cubic homogeneous map, combined with Zhao’s theorem, gives an SIC(22) witness. Thompson’s 24-variable and Harrison’s 79-variable cubic models similarly imply larger witnesses, whether or not those corollaries were written out as pairs.

The contribution here is narrower: removing the homogenizing variable lowers the explicit dimension from 22 to 21. The resulting linear  $BU$  block requires Lemma 1. At the audit cutoff, no earlier public  $SIC(n)$  witness with  $n \leq 21$ , or the nonhomogeneous lemma in the form used here, was located.

This is evidence, not proof, of priority. Public work is changing hourly and private or unindexed work may exist. We claim neither minimality of dimension nor minimality of support.

## References

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